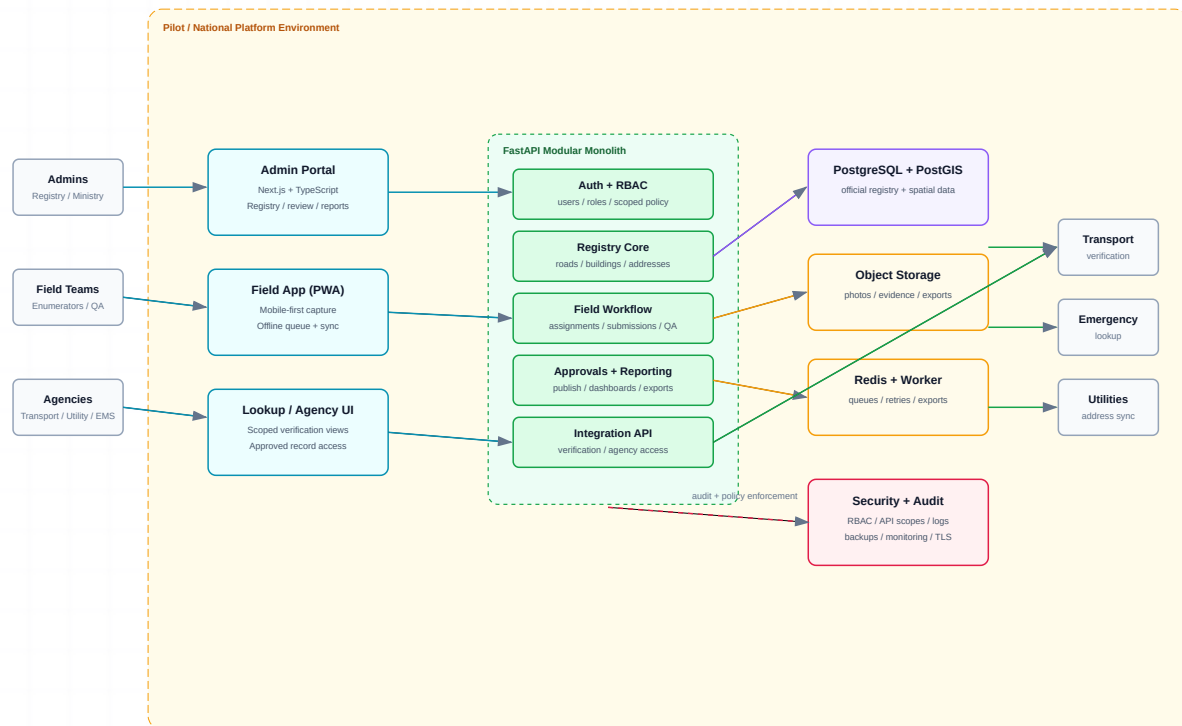


National Digital Addressing Platform Architecture

Government-owned address registry, field operations, approvals, and institutional verification stack



This first page keeps the platform flow visual: users enter through admin, field, or agency surfaces; the FastAPI core handles registry and workflow logic; Postgres/PostGIS, storage, queueing, and security support the system underneath.

Architecture Summary — Mobile Readable Version

Application Layer

What users actually touch

- Admin portal for registry administrators, ministry reviewers, validators, and helpdesk users.
- Mobile-first field app for enumerators and supervisors doing capture and QA in the territory.
- Scoped lookup and verification views for approved institutional users.
- Shared login and role scoping across all surfaces.

Core Platform Services

What the backend owns

- FastAPI modular monolith for the pilot, so the system stays operationally simpler early on.
- Registry core handles roads, buildings, addresses, versions, and publication states.
- Field workflow handles assignments, submissions, review, correction, and approval transitions.
- Reporting and integration APIs expose approved information safely to leadership and partner agencies.

Data Foundation

Where truth lives

- PostgreSQL + PostGIS is the source of truth for official registry and spatial records.
- Object storage holds photos, evidence files, and export artifacts.
- Redis + worker processes handle retries, queue jobs, exports, and async processing.
- Versioning and audit trails keep the platform governable, not just searchable.

Institutional Consumption

How other agencies use it

- Transport, emergency, and utility systems should consume verified address data through scoped APIs.
- External consumers should not get broad unrestricted registry access.
- Verification endpoints and approved exports are the safe first integration model.
- Broader interoperability can expand after pilot controls are proven.

Security & Operations**What keeps the system trustworthy**

- Role-based access, scoped permissions, and audit logging are mandatory.
- Backups, monitoring, TLS, and controlled releases are part of the platform, not optional add-ons.
- The pilot should run in a disciplined environment with staging and production separation.
- Start pilot-ready now, then harden and scale later without rewriting the whole architecture.

Plain-English version: this is a government-owned registry platform with field capture, approvals, official publication, and controlled agency verification — backed by a spatial database, object storage, background workers, and strict access control.